SOFTWARE ARCHITECTURE

Service-Oriented Architecture (SOA)

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- SOA modeling
- SOA pivotal components
- Types of SOA services

SOA definition

- SOA is an architectural style that primarily promotes service orientation.
- Service orientation implies **loosely-coupled systems** that are fundamentally focused on satisfying the business functions.
- It is easier to look at SOA as the solution to **interfaces and integration** problems, but it provides much more.
- It defines the vision and approach to enterprise architecture that builds upon software services representing heterogeneous business functions.
- SOA also mandates setting up operational cycles for running healthy systems as well as for maintenance and upgrade cycles.

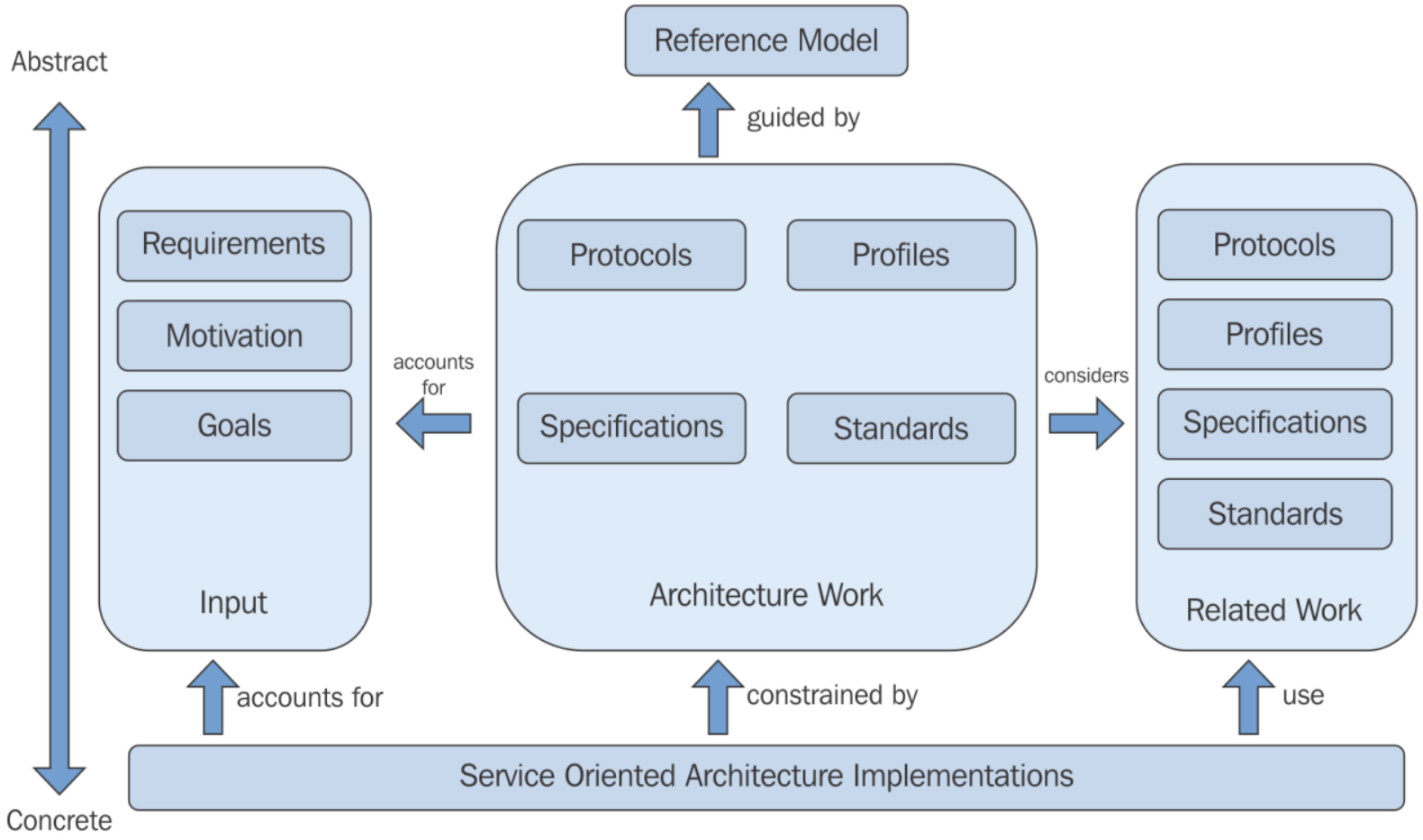
SOA modeling

- There are several SOA models and frameworks that exist in the market, but we will consider those which are the most referenced ones
- SOA reference Model
  - A reference model is an abstract framework to understand and describe the significant entities and **relationships** among them for some environments.
  - It is used or referred to for development of consistent standards or specifications supporting the environment of the enterprise.
  - The SOA Reference Model (SOA-RM) is an abstract framework that provides the fundamental concept of SOA to understand the entities and relationships between them, and for the development of SOA standards within an organization.

Principal concepts of SOA

- **Service**: A mechanism to enable access to one or more capabilities
- **Visibility**: A service provider and the service consumer need to see each other interaction with services: Performing actions against a service such as sending/receiving messages
- **Real world effect**: The intent and result of interacting with a service
- **Service description**: The information required information in order to interact with a service
- **Policies and contracts**: Policies are constraints to use a service, and contracts are an agreement between two or more parties
- **Execution context**: A set of infrastructure elements, process entities, policy assertions and agreements, which are identified as a part of service interaction between the service provider and service consumer

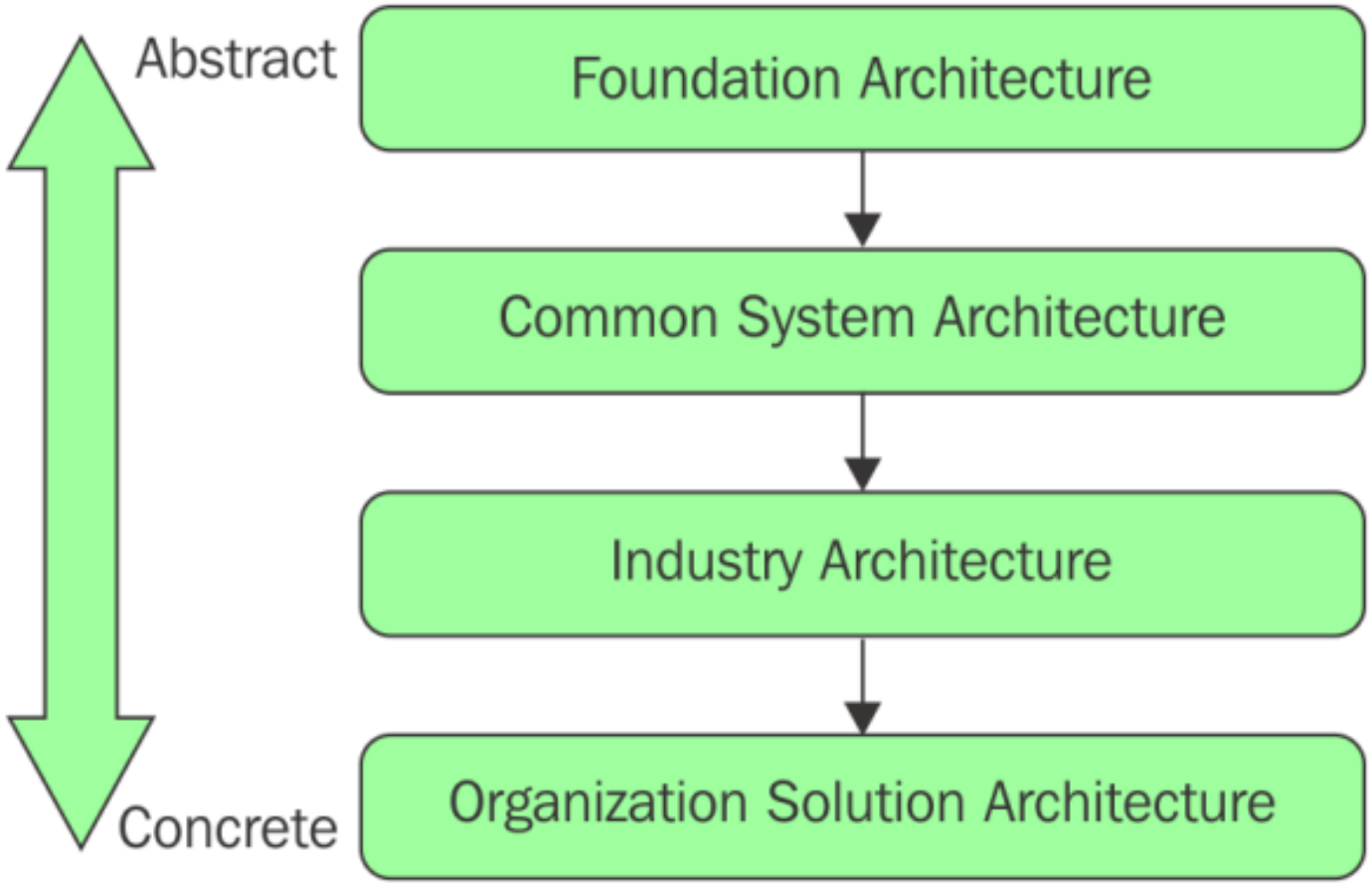
Reference model relationship



SOA Reference Architecture

- A reference architecture is an architectural design pattern that indicates how an abstract set of entities and relationships realizes a predetermined set of requirements, fundamental concepts, and relationships (that is, reference model) in the domain of interest.
- The purpose of reference architecture is to give a realization, which is a high-level solution artifact, and to architects, a standard model from which to architect specific solutions for their respective business domain.

SOA Reference Architecture



Reference model and reference architecture

- The relationship between a reference model and reference architecture has been defined very nicely by the SOA model
- Reference models work as the **fundamental guiding principles** for a given domain and are **more abstract** building blocks of the architecture models.

High-level layers in SOA (1/2)

- **Operational Systems layer**: This contains the operational system components of an enterprise. It primarily includes the infrastructure to support SOA implementation.
- **Service Components layer**: This includes the software components, which provide implementation for the service.
- **Services layer**: This includes services with description, contracts, policies, and containers with service components.
- **Business Processes layer**: It's a collection of the business processes of an enterprise.
- **Consumer Interfaces layer**: This includes the software components, which enable users to interact and use the services.

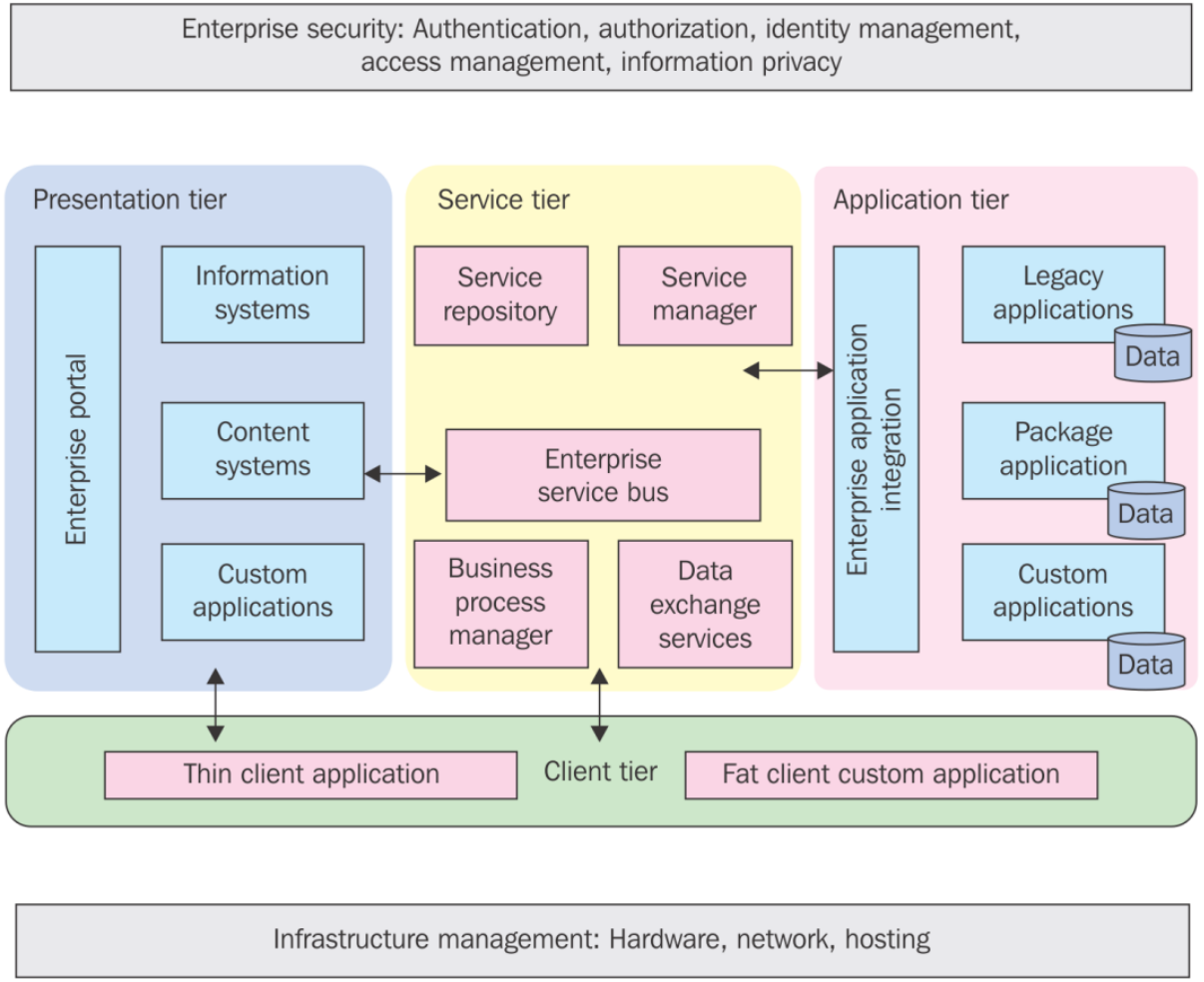
High-level layers in SOA (2/2)

- **Integration layer**: This is probably the most critical layer for an SOA platform. It provides the integration building blocks, including messaging, message transformation, message hub, event processing, service composition, and service registry discovery.
- **Quality of Service layer**: This layer basically monitors, reports (and manages) the quality of service of the SOA implementation, including its errors, performance, reliability, availability, and security.
- **Information layer**: This logically includes data, meta-data, analysis, interpretation, and transformation of data.
- **Governance layer**: This layer contains the governance rules and regulations, and their application on the services and operations.

Common reference information Architecture

- Information architecture basically shows the information flow in the SOA implementation environment, and the interactions between different tiers, systems, and processes.
- The following diagram shows the widely adopted IBM SOA Foundation reference information architecture

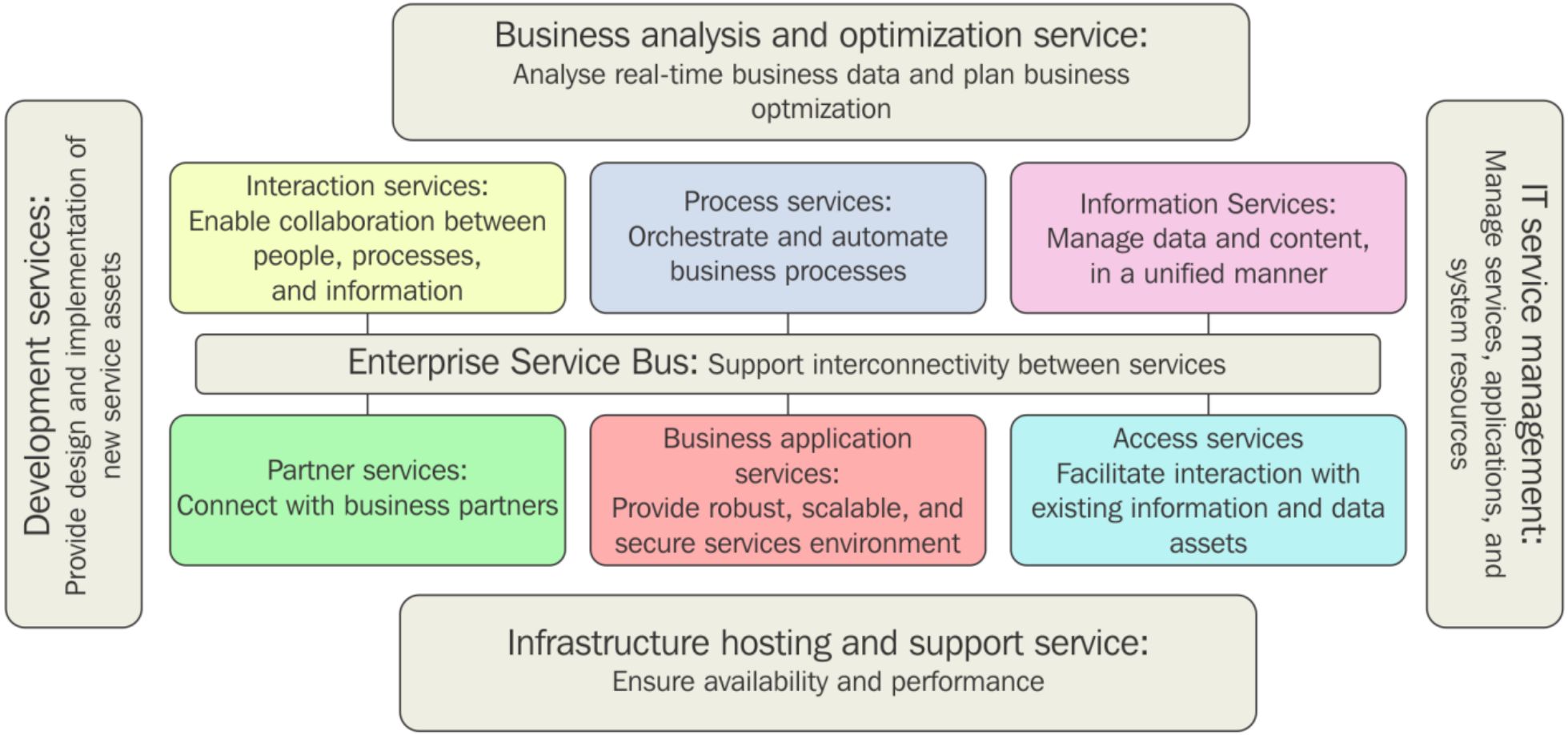
Common reference information Architecture



Common Reference Infrastructure Architecture

- The Infrastructure architecture depicts the fundamental infrastructure services required to **set up, build, deploy, and maintain** the SOA platform components.
- This helps in identifying the existing infrastructure, which could be affected by the SOA implementation. It is also useful when making **platform, technology, and vendor choices** for the SOA platform.

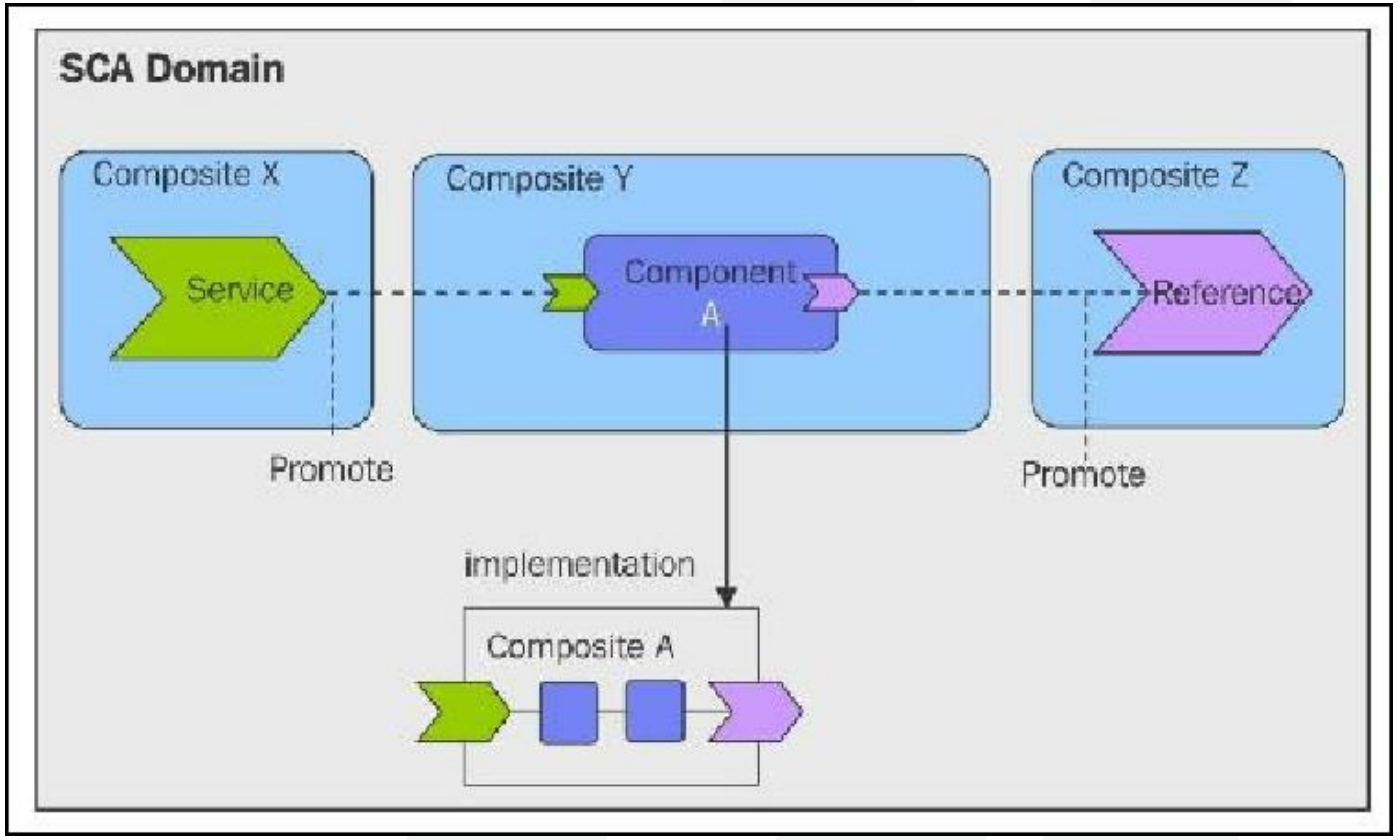
Reference Infrastructure Architecture (IBM)



Service Component Architecture

- Service Component Architecture (SCA) is one of the most popular standards, dating from 2005.
- It defines the architectural components of an SOA platform in a technical manner.
- It is standardized by the OASIS, mainly in collaboration with IBM, Oracle, and TIBCO.

Service Component Architecture



Service types

- Services are the most important aspect of the SOA.
- In an SOA platform, you have many services, and it is important to categorize them so that it is easy for service development and service management.
- By achieving business processes in terms of service, it increases the modularity, business understanding, improved and trackable information flow, and better organization of the business functionality; it also promotes reusability by exposing service interfaces to other services and systems.

The categories of SOA services (1/2)

Service categories	Short description
Interaction services	Primarily related to the presentation layer, it supports interaction between applications and users. Usually a part of the MVC pattern.
Process services	Also called composite services, they provide the composition logic, especially business flows. Could be part of the MVC pattern.
Information services	Expose the data logic of the enterprise design. Provide access to the data storage. Could be part of the MVC pattern as well. These services could be sub-divided and can have their own sub architecture with respect to the enterprise data architecture.
Access or adapter services	Provide access to legacy data or application. Provide access to a resource or expose the resources as a service.
Mediation services	They normally bind one service (usually, a producer) to more than one service (data consumers). They provide routing logic of data and services, data enrichment, and filtering. Some also call it a distributor service.
Business services	Provide encapsulated business functionality to internal and external consumers.
Security services	Enable application and enforcement of security attributes to other SOA services. Provide interactive and non-interactive layers of services as well as hookable services.

The categories of SOA services (2/2)

Service categories	Short description
Registry services	Enable service discovery and service registry as well as policies by providing managed access to various SOA artifacts.
Infrastructure services	Provide measurement and monitoring of infrastructure ensuring the integrity of the SOA operational environment.
Management services	Provide metrics, measurements, reporting services regarding other SOA services. Include outages, severe error detections and alerts, enforcing administrative policies towards achieving and maintaining the service level.
Development services	These are, generally, an entire suite of services related to tooling like modeling, development, testing, logging, debugging, instrumentation, error reporting, and alerting used in an SOA solution.
Strategy services	Mostly related to business intelligence services for improving business outcomes. Also may include the services that process the strategies of business to create an implementation roadmap covering both business and IT.

Service composition

- Service composition is the **combination** of existing services to create a new service, or to achieve a newer functionality.
- Since services are essentially business functions or processes in SOA, it describes the capabilities of business. SOA enhances the visibility and capabilities of a business by allowing them to create new functions or capabilities by combining the existing ones into a well-formed composition.
- There are two primary styles of service composition on a theoretical level, which are as follows
  - Service orchestration
  - Service choreography

Service orchestration

- Orchestration is defined as the arrangement of a musical composition for performance by an orchestra.
- Orchestration is a composition of services in which one of the services is a master service, which controls, schedules, and directs all the other services.

Service choreography

- Choreography as follows:
  - The art of representing dancing
  - The composition and arrangement of dances especially for ballet
- Basically, services can be compared to a set of dancers, and a business process to choreography, a direction, or a business process. Each dancer, or a group of dancers, or each service is autonomous in how they all carry out the direction.
- So, in service choreography, the composed services interact and cooperate with each other without the presence of a master, director, or controller service composing or combining them.

Common technology standards

- **Hypertext Transfer Protocol** (HTTP)
- **Extensible Markup Language** (XML): For data representation, transfer, and/or storage) SOAP (Messaging standard protocol)
- **Web Services Description Language** (WSDL)
- **JavaScript Object Notation** (JSON) has lately become more popular than XML) XPath, XQuery, and XSL Transformation
- **Business Process Execution Language** (BPEL) : XML- based construct for services orchestration)
- **Universal Description, Discovery, and Integration** (UDDI)

Service discovery

- One may **hardcode** the URL of a service in the service consumer configurations, or one can use the **dynamic mechanism to fetch the URL** at program execution (initialization) time. This fetching of the actual URL is part of service discovery.
- UDDI is one of the ways to register services and their URLs, and to retrieve them by their respective consumers. Service discovery mechanism can also be custom-built by an organization in combination with the configuration management system. Most of the SOA solutions provide service discovery as part of the solution.
- This feature of SOA does not necessarily enforce the policy framework, while some of the service discovery solutions offer lifecycle management as well.

Enterprise Service Bus (ESB)

- Enterprise service bus or ESB is a **collection** of features, functionality, and tooling, and has become almost an essential component, rather a backbone, for any SOA implementation.

ESB Segments

- Storage resources (in memory, persistent, transfer)
- Gateway services (protocols, security)
- Message broker (as explained previously)
- Solution designer (IDE)
- Service adapters (data and services connectors to various resources including third parties)
- Management Interface (administration, installation of components, management, and monitoring)

ESB features

- Location transparency (producer and consumers are unaware of the endpoints)
- Messaging
- VETRO
  - Validation (format and data validation)
  - Enrichment (addition of attributes)
  - Transformation (conversion from one format to another)
  - Routing (for example, message dispatching based on filters)
- Process and message monitoring
- Message security
- Protocol conversion (for example, HTTP to JMS)
- Monitoring, Administration, and Tracking

Data modeling

- Data modeling, including business and data architecture, is part of SOA architecture, which is a big topic by itself.
- We would just mention here some of the essential things with reference to the data modeling for SOA implementation.

Master Data Management (MDM)

- MDM includes information, processes (creation and maintenance), policies, standards, governance, and tools common across the enterprise.
- It holds these two main types of data:
  - Reference data (data referenced across the organizational system)
  - Analytical data:
    - Identification and verification of entities and their relationships
    - Analytics on master data, for example, number of total customers increased in a month
    - Analytical information (for example, in terms of metrics) fed from data warehouse(s)

Common data model

- This is also a part of data architecture, which, basically, describes the data model to be used across the span of SOA-wide services.
- The common data model is defined and synchronized between persistent storage (database) and the message schemas travelling throughout the SOA services.
- Common data model also includes transformations to and from various other data models depending on the outward and inward interfaces from systems outside SOA or legacy systems.
- Transformations can be dynamic at runtime, ETL scheduled, or could be offline, triggered by external processes in a batch conversion manner.

Services gateway

- SOA platforms usually (but not necessarily) have a services gateway.
- These gateways are meant to have most of the web services registered inside them for controlling access, throttling, statistical measurements, QoS, and service mappings besides other functionalities.

Main features of services gateway

- HTTP and JMS-based service producers and consumers
- Security (authentication, authorization, certificates, encryption, and so on)
- Validation (for example, WSDL, SOAP Requests, and more)
- Throttling
- Routing (web services routing) Service Proxy
- Transformation (for example, from one service to another)
- Caching
- Central logging and reporting (real time and non-real time)
- Mediation (becoming common across gateways)

SOA services library

- SOA services library is not an essential component of an SOA platform, but these kind of libraries are common to SOA implementations.
- It is an enterprise library, generally used within implementation technologies such as .NET, Java, and various others.
- The purpose of this library is to encapsulate the implementations of the most common concepts such as logging, which are used across (almost) all of the implementations of SOA services.

The concepts in SOA services library

- Common implementation technology framework libraries
- Context (SOA service meta business context)
- Logging
- Error handling mechanism
- Common services structure
  - Constants
  - Common schemas, XML, JSON, and Strings Service or system states
  - Common SOA internal service references (static and dynamic)
- Information to enable caching mechanism in SOA

Tracking, logging, and error handling in SOA

- **Tracking mechanism** enables tracking of input message and message processing at various stages, which often span across multiple services.
- **Logging mechanism** is a general logging feature of any SOA service implementation, which is unified so that all services log in a given SOA standard.
- **Error handling mechanism** deals with error logging, related severity, priority, and handling of error in terms of reporting or retrying/correction. Error reporting may include monitoring and alerting services.

The attributes that help facilitate mechanisms

- Log Levels (uniform log levels for all implementations)
- Timestamp (includes time zones for global enterprise)
- RequestID (GUID generated by the Sender/Client application to track the transactional flow)
- TrackingID (GUID generated to keep track of message flow) BusinessUnits (inserted by the Application)
- DomainID, ApplicationID, ModuleName (these are pre-defined values, perhaps in SOA services library)

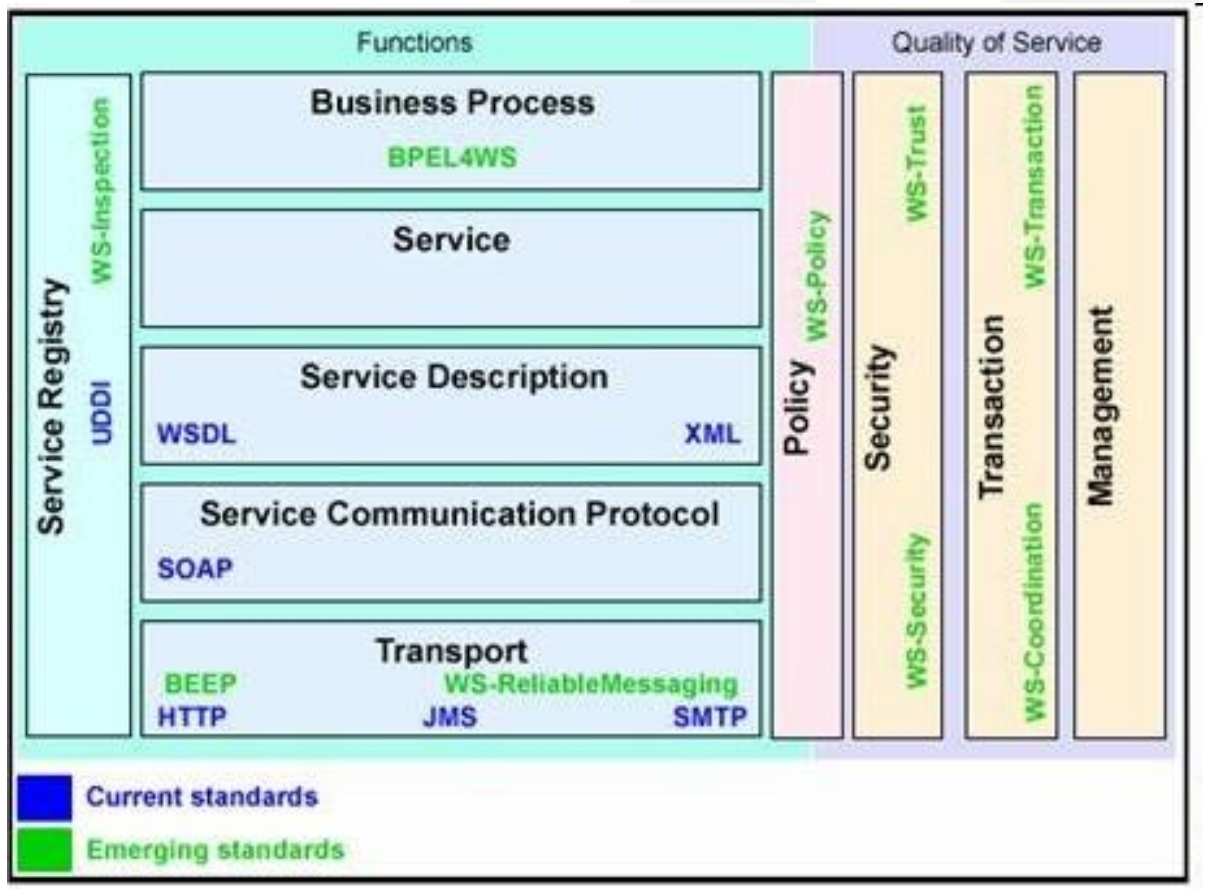
The attributes that help facilitate mechanisms

- ServiceStep (used by Mediation / Orchestration)
- BusinessKey (context sensitive info for a message or a set of related messages)
- RetryCount (retries indicator if error retry/correction is activated)
- MessageType (Pre-defined values to identify the type of a business message)
- Message (could be short message and a full message)
- Description (useful for logging)

SOAP Service

- First introduced in 1999
- Simple Object Access Protocol (SOAP) is a standards-based Web services access protocol that has been around for a while and enjoys all of the benefits of long-term use.
- Originally developed by Microsoft, SOAP really isn't as simple as the acronym would suggest.

SOAP Service structure



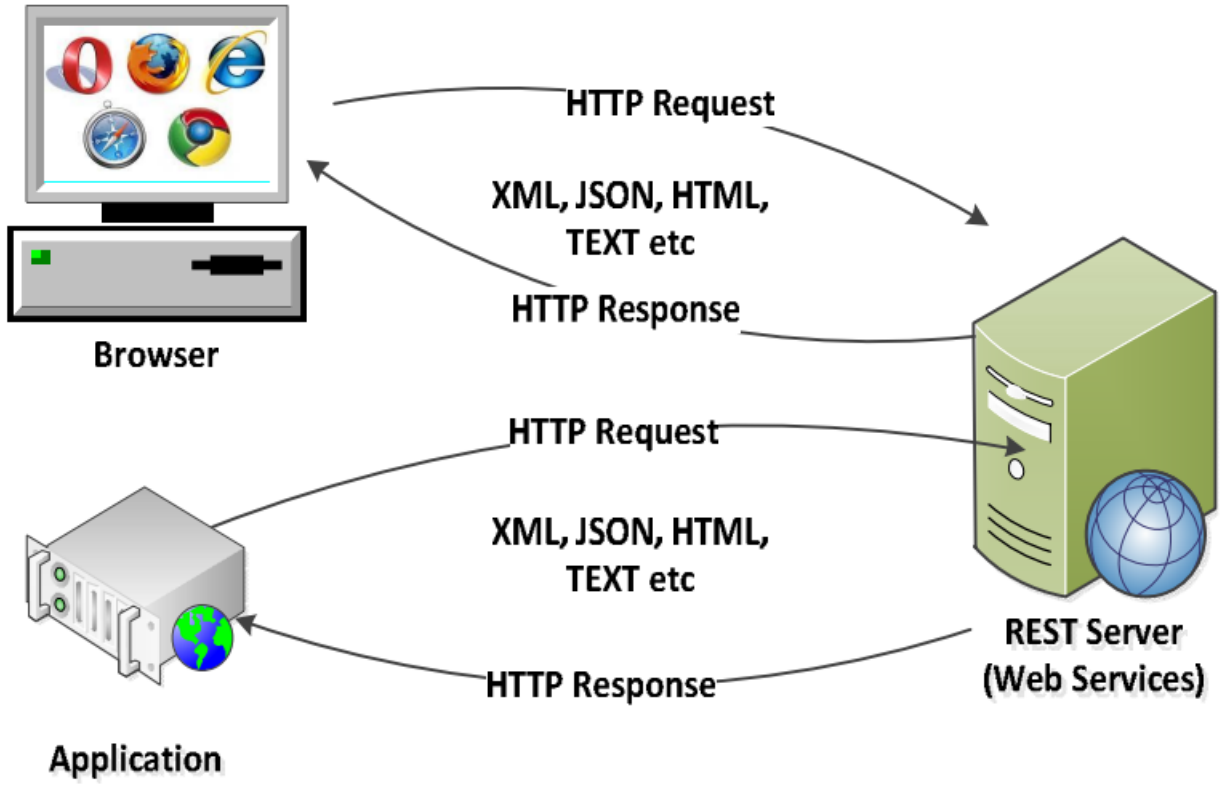
REST Service

- First introduced in 2000
- Representational State Transfer (REST) is a technology that relies on a stateless, client-server, cacheable communications technology that uses the HTTP protocol.
- REST is considered an architectural style for developing applications that communicate over the network.
- REST uses HTTP for all four CRUD (Create/Read/Update/Delete) operations, so it is a lightweight and simple alternative to the complex mechanisms stated above.

RESTful

- RESTful is a lightweight Web Service:
  - Platform-independent (don't care whether the server is Unix, the client is MacOS, or Windows),
  - Language-independent (C# can talk to Java, ...),
  - Standards-based (runs on top of HTTP, XML, JSON, …),
  - Easily used in the presence of firewalls, as it is simply uses the popular HTTP protocol that can bypass any firewalls used.
- RESTful applications use HTTP requests to post data (create and/or update), read data (like queries), and delete data.

RESTful communication



Summary

- To understand and build the SOA software implementation architecture.
- Define architecture to SOA-specific architectural concepts using the right aspects and attributes to consider while modeling the SOA platform.
- The essential artifacts required for developing a general SOA architecture and gradually moved towards the technical perspective of the implementation architecture.

Q&A

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